ET AUTOTECH HACKATHON 2026 | IDEA SUBMISSION

NEURAL-SHIELD

Edge AI Predictive Maintenance for Resilient Smart Manufacturing

A $5 microcontroller that learns a machine's healthy vibration signature on-device, cuts power 4 milliseconds after a catastrophic fault, and auto-drafts the replacement purchase order.

THEME CHOSEN

Theme 1 - AI for Resilient Automotive Supply Chains & Smart Manufacturing

TEAM NAME

Aryan Gupta

TEAM MEMBERS

Aryan Gupta

01 / THE PROBLEM

Unplanned Downtime Breaks the Production Line

Automotive plants run on legacy heavy machinery - spindles, presses, conveyors. Replacing them costs millions, so they stay. When one fails unexpectedly, the entire line stops, deliveries slip, and downstream suppliers stall.

- **Cloud is too slow for physics.** Standard answer is cloud monitoring: stream sensors up, analyze, send a stop signal back.
- **Failure outruns the network.** A slipping gear or seizing bearing cascades into destruction in milliseconds. A 200 ms - 2 s round trip arrives after the machine is already wrecked.
- **Emerging-market constraints.** Indian factory floors face patchy connectivity, brownouts, and no budget for $1000+ gateways per machine - cloud-first CBM simply does not deploy there.
- **The recovery gap.** Even after a correct shutdown, repairs stall: diagnosing the part, checking stock, choosing a supplier, and raising a purchase order takes hours to days - all downtime.

Detect-to-Stop: Today vs NEURAL-SHIELD

Approach	Latency	Offline-safe	Cost/node
Cloud monitoring	200 ms - 2 s+	No	$$$ + subscription
Manual inspection	Hours - days	Yes	Skilled labor
NEURAL-SHIELD	~4 ms	Yes	~$5 (ESP32)

$4,200
DOWNTIME COST PER HOUR (ONE LINE)

50,000x
FASTER THAN A CLOUD ROUND TRIP

~$5
HARDWARE COST PER PROTECTED MACHINE

02 / PROPOSED SOLUTION

Give Old Machines a Fast, Local Brain

THEME 1: RESILIENT SUPPLY CHAINS & SMART MANUFACTURING

NEW SOLUTION - NOT AN INCREMENT

NEURAL-SHIELD retrofits legacy machinery instead of replacing it. A 1.58-bit ternary neural network is compressed to fit entirely inside a commodity ESP32. The safety decision never leaves the chip; the supply-chain response is automated the second the machine stops.

1. LISTEN

Auto-calibration

5-second warm-up learns this specific motor's healthy vibration distribution. Thresholds self-tune at mean +3 sigma / +6 sigma - zero manual configuration.

2. PREDICT

On-device ternary NN

Forecasts the next vibration window from the last 200 samples; prediction error (MSE) measures deviation from the learned healthy signature.

3. ACT

4 ms hardware latch

Confirmed fault trips a hardware latch that severs motor power directly - no network in the loop. Machine coasts to a safe stop before damage spreads.

4. RECOVER

Agentic procurement

A local Llama-3 agent reads live fault data + factory inventory, identifies the failed bearing, picks the best supplier, and drafts the purchase order.

Why new: existing CBM products improve cloud analytics. NEURAL-SHIELD inverts the architecture - sub-5 ms reflex on a $5 chip, plus an autonomous procurement agent closing the loop from fault to purchase order.

LAYER 1 - 1.58-bit Ternary Neural Network (on-chip)

- **Weights:** Every weight constrained to {-1, 0, +1} - the 1.58-bit regime (log2 of 3). Multiplies become add / subtract / skip: no floating-point matmul at inference.
- **Architecture:** 200 -> 1024 -> 512 -> 256 -> 10 fully-connected, GELU. Reads 200 vibration samples, forecasts the next 10.
- **Footprint:** 864K parameters packed at 2 bits/weight = 215 KB - fits in ESP32 SRAM/flash. Full forward pass in ~4 ms on a 240 MHz core.
- **Anomaly = prediction error:** Trained only on healthy vibration (CWRU baseline). Healthy machine = predictable = low MSE. Any fault mode - even unseen ones - spikes the prediction error. Unsupervised by design.

```
MSE = mean( (actual_window - predicted_window)^2 )
```

one number, computed on-chip, drives every decision below

LAYER 2 - Four-Pillar Industrial Gatekeeper

- **Auto-calibration:** warning = mean + 3 sigma, critical = mean + 6 sigma - learned per machine at boot
- **EMA smoother:** alpha = 0.15 filters transient noise out of the raw MSE stream
- **Persistence gate:** critical must hold for 5 consecutive windows - a glitch cannot stop the line
- **Hardware latch:** confirmed fault locks power off until physical reset / supervisor override

LAYER 3 - Autonomous Logistics Agent (Llama 3)

- On a stable -> critical trip, the bridge fires the agent with live MSE + factory context: inventory levels, supplier database, downtime cost.
- Agent reasons over lead time vs price vs stock-outs, drafts the fault report and a supplier purchase order in seconds.
- Deterministic fallback guarantees a valid PO even with the LLM offline - the demo never blocks.

04 / TECH STACK

Proposed Tech Stack

EDGE FIRMWARE

ESP32 DevKit (~$5)

- C++ / Arduino core
- Ternary inference engine (custom, 2-bit packed weights)
- 4-pillar gatekeeper + hardware latch
- WebSocket telemetry over WiFi

MODEL & TRAINING

PyTorch

- Ternary quantization-aware training
- CWRU bearing vibration dataset (healthy baseline)
- Exports packed C header (ternary_weights.h)
- 200->1024->512->256->10, GELU

REALTIME BRIDGE

Python

- websockets server (:8000) - single source of truth
- Status classifier + agent trigger (debounced)
- Relays dashboard commands to the edge node

DASHBOARD

Next.js 15

- React 19 + Tailwind CSS 4 + Framer Motion
- Physical Digital Twin (code-generated SVG spindle)
- Live waveform, 3-state system, agent terminal
- Fully frozen when the bridge disconnects - zero fake data

AI AGENT

Llama 3 via Ollama

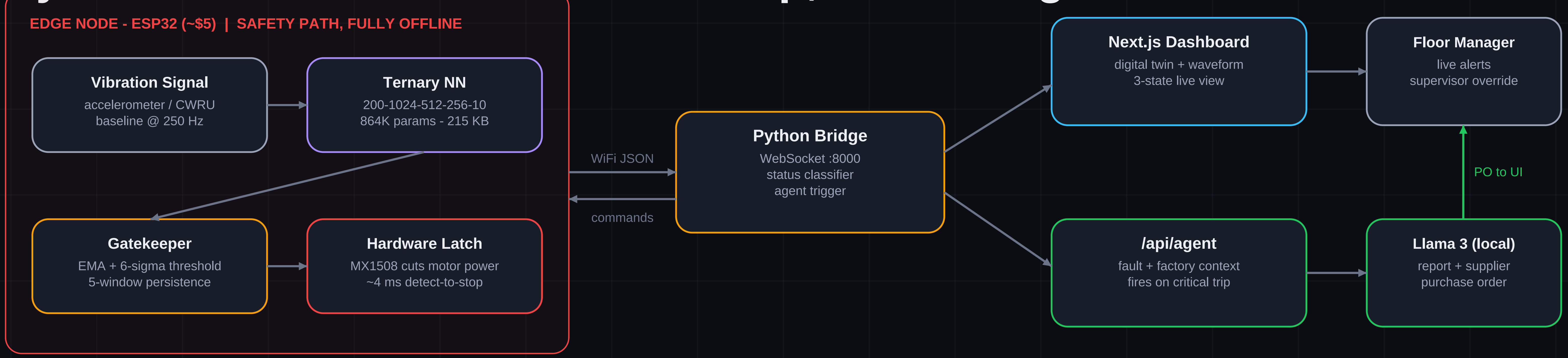
- Runs 100% locally - no cloud API
- Factory context DB: inventory + supplier matrix
- Drafts fault report + purchase order
- Deterministic fallback path

HARDWARE

Actuation chain

- MX1508 H-bridge - motor power switching
- 12 V spindle motor (machine under protection)
- Buck converter 12V->5V, common-ground design
- Onboard LED status indicator (GPIO2)

System Architecture: Reflex On-Chip, Reasoning On-Prem



DESIGN PRINCIPLE - THE SAFETY LOOP NEVER LEAVES THE CHIP

Vibration -> inference -> latch executes entirely on the ESP32. The bridge, dashboard, and agent are observability and logistics: if WiFi, the server, or the internet dies, the machine is still protected. This is what makes the system deployable in low-connectivity Indian plants where cloud-first solutions fail.

06 / WORKING PROTOTYPE

Built and Running - Demo Evidence

This is not a concept: the full pipeline - firmware, bridge, dashboard, agent, and physical actuation - is implemented and demonstrated live.

[ADD SCREENSHOT: dashboard in STABLE state - digital twin spinning, live waveform]

[ADD SCREENSHOT: CRITICAL trip - relay badge + AI agent terminal drafting the purchase order]

[ADD PHOTO: hardware rig - ESP32 + MX1508 + 12V spindle motor]

SOURCE CODE - PUBLIC REPOSITORY

github.com/aryxnsdfs/NEURAL-SHIELD-

Full firmware, training pipeline, bridge, dashboard, and agent - MIT licensed, reproducible end-to-end.

[ADD LINK: demonstration video URL]

BUSINESS IMPACT MODEL

- **ROI:** One line-down hour costs $4,200. One averted catastrophic failure pays for ~840 NEURAL-SHIELD nodes.
- **Recovery time:** Fault-to-purchase-order drops from hours/days of manual diagnosis and procurement to under a minute, automatically.
- **No false stops:** Warning state keeps production running on minor anomalies - alerts the floor manager without stopping the line.
- **Capex avoided:** Retrofit, don't replace: protects millions in legacy capital equipment for $5 per machine.

SCALABILITY

- **Fleet-ready:** Auto-calibration means zero per-machine engineering: flash the same firmware on any motor, press, or pump - it learns its own baseline in 5 seconds.
- **Emerging-market fit:** No cloud dependency, no gateway, no subscription - scales to thousands of nodes in connectivity-poor plants across India.
- **Generalizes:** Same prediction-error method extends to any vibrating asset: CNC, conveyors, compressors, EV drivetrain test rigs.

FEASIBILITY - ALREADY PROVEN

Working hardware

Live demo rig: ESP32 + MX1508 + 12 V spindle. Inject a fault from the dashboard - the motor physically cuts and coasts to a stop, agent drafts the PO.

Real data, real model

Trained on the CWRU bearing dataset (industry-standard). 864K-param ternary model verified on-device at 215 KB - measured, not estimated.

Honest engineering

Dashboard shows zero fake data: every number streams from the bridge. Bridge offline = UI frozen. Judges see exactly what the hardware sees.

08 / WHY US

Why This Solution Must Be Considered

Depth of problem insight	Attacks the real physics constraint - failure propagates faster than any network round trip - and the recovery gap nobody automates: procurement.
Innovation & originality	First-principles inversion of cloud CBM: 1.58-bit ternary NN on a $5 chip + hardware latch + an agentic LLM closing the loop from fault to purchase order.
Intelligence architecture clarity	Three explicit layers - forecasting NN, statistical gatekeeper, reasoning agent - each with a defined job, interface, and failure mode.
Feasibility & technical soundness	Fully working prototype: real dataset, measured 215 KB on-device model, physical actuation, deterministic fallbacks at every layer.
Impact & scalability	$5/node, zero-config self-calibration, no cloud dependency - deployable today across thousands of machines in Indian plants.
Clarity of demo	Live end-to-end story in under 2 minutes: healthy -> warning -> critical -> power cut -> AI-drafted purchase order, all visible on one dashboard.

ROADMAP

- **Phase 1 (now)**
  Working prototype - vibration forecasting, hardware latch, agentic procurement, live dashboard.
- **Phase 2**
  MEMS accelerometer (live ADC) on production machinery; OTA model updates; multi-asset bridge.
- **Phase 3**
  ERP integration (SAP/Tally) for auto-raised POs; fleet dashboard; process-capability analytics (Cp/Cpk) from the same vibration stream.
- **Phase 4**
  Supplier-risk scoring on accumulated failure data - closing the full Theme-1 loop from shop floor to sourcing strategy.

AT A GLANCE

Model	1.58-bit ternary, 864K params, 215 KB packed
Inference	~4 ms per window on a 240 MHz ESP32 core
Detection	Unsupervised prediction-error (MSE) - no fault labels needed
Thresholds	Self-calibrated: mean + 3 sigma / + 6 sigma per machine
False-alarm guard	EMA + 5-window persistence + hardware latch
Agent	Llama 3 local via Ollama + deterministic fallback
License	MIT - fully open source

[ADD: team photo / contact email / phone]

[ADD: any awards, prior validation, or mentor feedback]

THANK YOU

NEURAL-SHIELD - Retrofit, don't replace.

Team Aryan Gupta | ET AutoTech Hackathon 2026
github.com/aryxnsdfs/NEURAL-SHIELD-